FIELD GOAL PROBABILITY MODEL

Predicting field goal success probabilities to support game decision-making and kicker evaluation.

Lauren Manis

MODEL PURPOSE & OVERVIEW

The goal of this model is to estimate the probability of field goal success in the NFL, leveraging key contextual and environmental factors such as distance, weather, game situation, and kicker performance. The predictions are designed to inform critical coaching and front office decisions, including whether to attempt a field goal or go for it on fourth down, evaluating kicker reliability, or making roster and contract calls.

The model is a **Logistic Regression built on football-intuitive features**—selected based on domain knowledge and practical relevance. This model was chosen for its strong balance of accuracy, interpretability, and real-world reliability, making it especially well-suited for high-leverage football decision-making.

MODEL DETAILS

The model was trained on all field goal attempts from all NFL regular and postseason games between 2010 and 2023 (N = 12,829), along with a small set of synthetic misses added to account for selection bias in long-distance kick attempts, and tested on all attempts from the 2024 season.

- Data was sourced from the nflfastR package, which provides detailed play-by-play tracking for every NFL game.

An initial baseline model - a simple logistic regression using only kick distance - achieved an AUC of 0.774. The final model builds on this by adding a small, targeted set of football-intuitive contextual features, recognizing that distance remains the primary driver of field goal success.

- Only features that are consistently available, interpretable, and impactful were included to ensure simplicity and generalizability.

Features

Kick Distance (yards)	**Wind** (mph)	**Adjusted FG%**
Season	**Precipitation** (Binary)	**Rookie** (Binary)

Kicker FG%: cumulative within-season overall FG%, regressed to the league average.
Last 2 Minutes: 1 if the kick took place within the final 2 minutes of the 1^{st} or 2^{nd} half of the game.

ACCURACY METRICS

AUC, Precision, and Log Loss were prioritized as the key evaluation metrics

- **AUC**: Measures overall classification performance.
- **Precision**: Emphasized because false positives are riskier in coaching decisions—better to err on the side of caution.
- **Log Loss**: Penalizes overconfident incorrect predictions, helps assess calibration.

AUC	ACCURACY	PRECISION	LOG LOSS
0.74	0.80	0.88	0.38
Indicates a strong ability to distinguish between made and missed field goal attempts.	82% of all field goal attempts were correctly classified by the model.	89% of predicted made field goals were actually made.	Indicates well-calibrated predictions, predicted probabilities are close to actual frequencies.

Metrics shown are based on 2024 test set performance

Note: Accuracy is reported for completeness, though less informative in imbalanced datasets

MODEL SELECTION

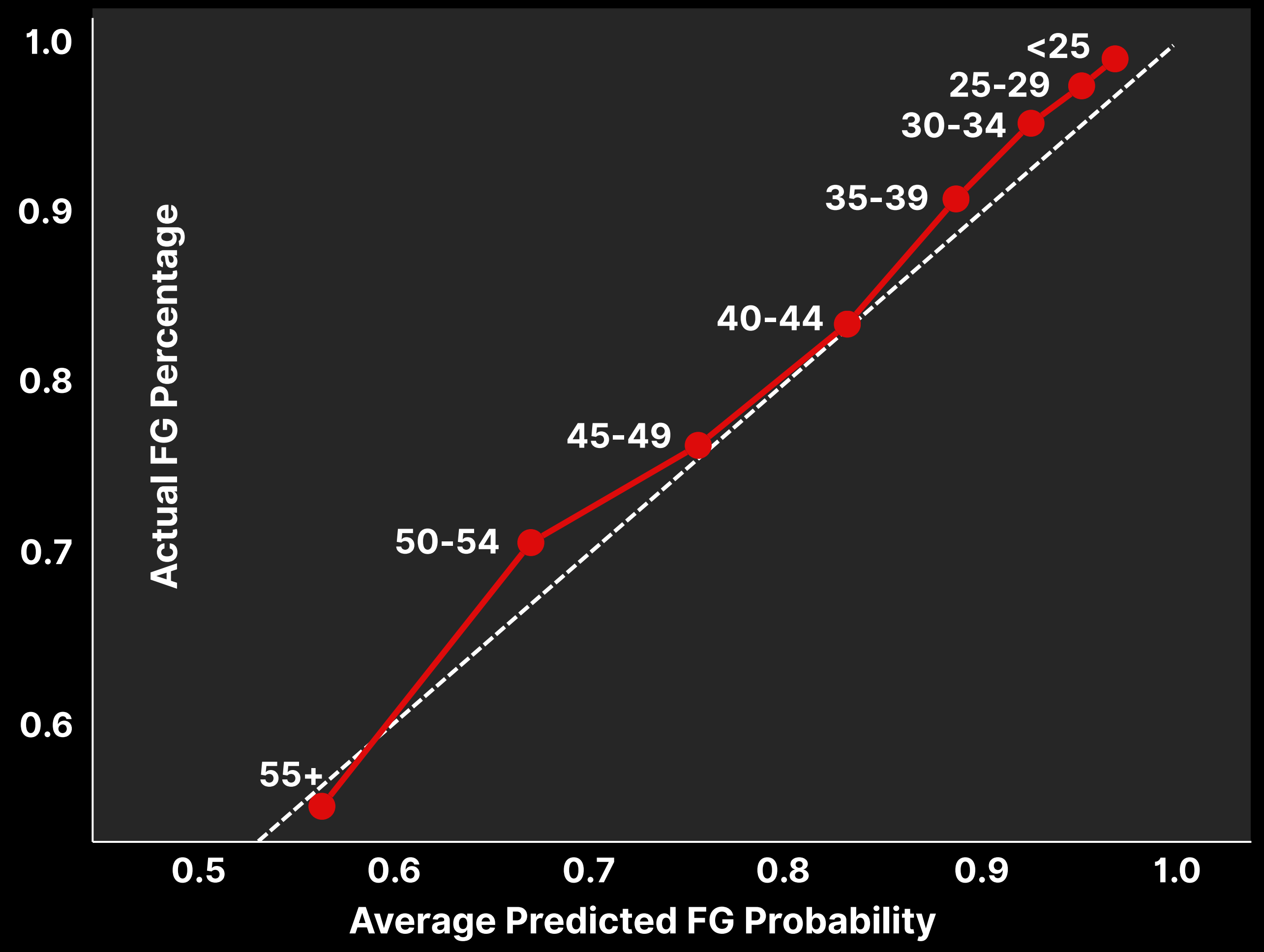
A total of 8 models were built and evaluated, including logistic regression, lasso, random forest, and weighted variants.

Ultimately, the **Logistic Regression with Football-Intuitive Features** (“Football Model”) model provided the best balance of performance, calibration, and interpretability.

- This specific “Football” model matched or outperformed more complex alternatives while remaining simple, transparent, and coach-friendly.

Predicted Vs Actual FG% by 5-Yard Bucket

Football Model, All Data



MODEL ASSUMPTIONS

- Blocked field goals were excluded from the dataset
  - They account for <2% of observations and tend to behave like noisy misses, so were excluded for clarity.
- Snaps and holds were not accounted for, as there is no reliable data source to identify botched snaps/holds.
- Kick direction (left/right hash) is not included due to lack of standardized data.
- False positives are more costly than false negatives, given coaching preference for conservative decisions.
- Kicker FG% inputs are cumulative within-season stats, including only attempts prior to each kick (no data leakage).
- Adjusted FG% is regressed to the league average, especially early in the season, to stabilize estimates and account for sample size.
- Model trained on attempts from 2010–2023, excluding earlier years due to significant changes in league-wide field goal success rates.
- Synthetic misses from 55+ yards were injected to address sample bias.
  - Coaches are less likely to attempt long kicks with weaker kickers, which results in an upward bias in observed make rates at long distances.
  - To better reflect real-world outcomes, a small number of plausible long-distance miss scenarios were added by repurposing punts taken under favorable kicking conditions, with a placeholder kicker assigned per team.